Ricky Fan

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Education

McMaster University Hamilton, ON
Bachelor of Applied Science in Computer Science, GPA: 3.88/4.0 *Sept. 2019 – April 2024*

Experience

Software Developer June 2024 – Present
GEOTAB *Oakville, ON*

- Led the multi-region migration of the High Availability Gateway Registry, an Orleans service serving device metadata, moving cluster membership from Kubernetes etcd to Cloud Spanner and single-cluster ingress to a Cross-Region ALB with Istio mesh
- Built the supporting GCP infrastructure in Terraform (cross-region ALB, DNS geo-routing, VPC peering) and validated rollouts with k6 load tests scaled to 60 replicas
- Removed an N+1 Orleans grain activation pattern causing cluster-wide P99 latency spikes, replacing per-vehicle grain calls with one bulk Spanner query that cut a 1,000-vehicle cold request from 94s to 3.4s
- Re-architected message delivery in the telematics gateway from semaphore-based locking to a channel-based writer, eliminating production write timeouts on persistent device socket connections
- Decoupled the gateway from Geotab's fleet-management monolith by pruning transitive references across 20+ modules and containerizing it for independent deployment
- Built Grafana and Superset dashboards covering the Istio service mesh, registry cluster health, and per-stage data-download latency, isolating bottlenecks during on-call incidents

Software Developer Intern Sept 2022 – April 2023, Sept 2023 – May 2024
GEOTAB *Oakville, ON*

- Debugged and fixed parallel test failures caused by a mutex race condition in C#, reducing test failures by 98%
- Refactored Gateway's Protobuf message processor for maintainability and testability, achieving 100% coverage with mock/stub-based unit tests
- Developed an automated pipeline tool and a C# CLI (GatewayPiper) to deploy test rigs and run health checks using GitLab CI, Terraform, Kubernetes, Helm, and GKE

Software Developer Intern May 2023 – Aug 2023
BlackBerry QNX *Ottawa, ON*

- Developed an internal PyPI package leveraging the Jira and Jenkins APIs to automate the change-log request process for a safety-certified microkernel RTOS, using multithreading to service concurrent requests

Projects

spx | *Rust, Tokio, gRPC* Feb 2026 – Jun 2026

- Implemented a Multi-Paxos protocol in Rust, combining Raft's leader election with Spanner-style leader leases and TrueTime for external consistency
- Designed a four-role state machine (Follower, Pre-Candidate, Candidate, Leader) driven by a single tokio::select! loop multiplexing I/O, lease expiry, and heartbeat timers
- Built time-bounded leader leases and commit-wait logic to safely serve local reads without quorum checks

x86OS | *C, x86 Assembly* May 2023 – Sep 2023

- Built a 32-bit x86 OS from scratch, including a two-stage bootloader transitioning the CPU from 16-bit real mode to 32-bit protected mode
- Implemented core kernel subsystems (GDT, IDT, PIC, TSS) and interrupt handlers for page faults, syscalls, the PIT timer, keyboard, and floppy disk controller
- Developed a virtual memory manager with paging, a higher-half kernel, a DMA floppy disk driver, and a FAT12 file system

Technical Skills

Languages: C#, Rust, Python, Java, C/C++, x86 Assembly, Bash, JavaScript, SQL
Cloud & Infrastructure: GCP, Kubernetes (GKE), Terraform, Helm, ArgoCD, Istio, Docker, GitLab CI
Frameworks & Tools: .NET, Orleans, gRPC, Protobuf, Tokio, Spring Boot, React, Node.js, Cloud Spanner, PostgreSQL, Grafana, Prometheus, Superset, k6, Git